Tim Reska

AI-Augmented Medical Underwriting

What I built in 48 hours with Claude to show you what's possible

PhD Helmholtz Munich · github.com/ttmgr · March 2026

I discovered "plausible but wrong" in my own published research

Tim Reska | PhD Helmholtz Munich & TU Munich | github.com/ttmgr

RESEARCH & PUBLICATIONS

- **8 peer-reviewed publications** including **Nature Communications**
- 22,000+ accesses, 43+ citations
- Invited speaker: **ETH Zurich, Cambridge, TUM**
- PhD thesis: real-time genomics for pathogen detection

AI & CLAUDE EXPERTISE

- **36-month LLM evaluation**: 22 models (GPT, Claude, Gemini)
- Published the **"plausible but wrong" failure mode** (ISME Communications 2024)
- Integrated Claude + MCP into live sequencing pipelines (**-40% dev time**)
- **10 modular pipelines** deployed across 7 international sites in DE/FR/ES

> "I applied my PBW research to medical underwriting — and built the proof in 48 hours."

Underwriting models are confidently wrong when data quality is poor

I found this failure mode in genomics AI — it applies directly to insurance

THE FAILURE MODE

- Model scores patient as "78% high risk" based on 3 diagnoses, no labs
- Score **looks right** — passes every sanity check
- But it's echoing training data averages, not this patient's reality
- I call this **"Plausible but Wrong"** (ISME Communications 2024)

AT PORTFOLIO SCALE

- 100,000 decisions/year × 2% PBW = **2,000 mispriced policies**
- **Invisible** in standard metrics (AUC, Brier score look fine)
- EU AI Act 2024: Art. 14+15 require **per-case reliability assessment**
- No production system today validates this at individual level

I built a working underwriting system in 48 hours to show you what's possible

Not a slide deck — a running application with 231 tests and every number verified against PubMed

THE PIPELINE (SIMPLIFIED)

```
Patient Record → Data Quality Score → Right Model → P(wrong) → Accept / Review / Reject
```

WHAT IT DOES

- Scores data quality **before** the model runs
- Routes to the right model for available data (**no imputation**)
- Estimates **P(wrong)** and makes cost-optimal decisions
- Full **audit trail** for every decision (EU AI Act ready)

HOW I BUILT IT

- **Claude as co-pilot** for architecture, coding, and testing
- **LLM agents** for PubMed literature verification
- Same AI workflow I'd use at Allianz — but with real data

International markets show 2.4x higher mispricing risk than Germany

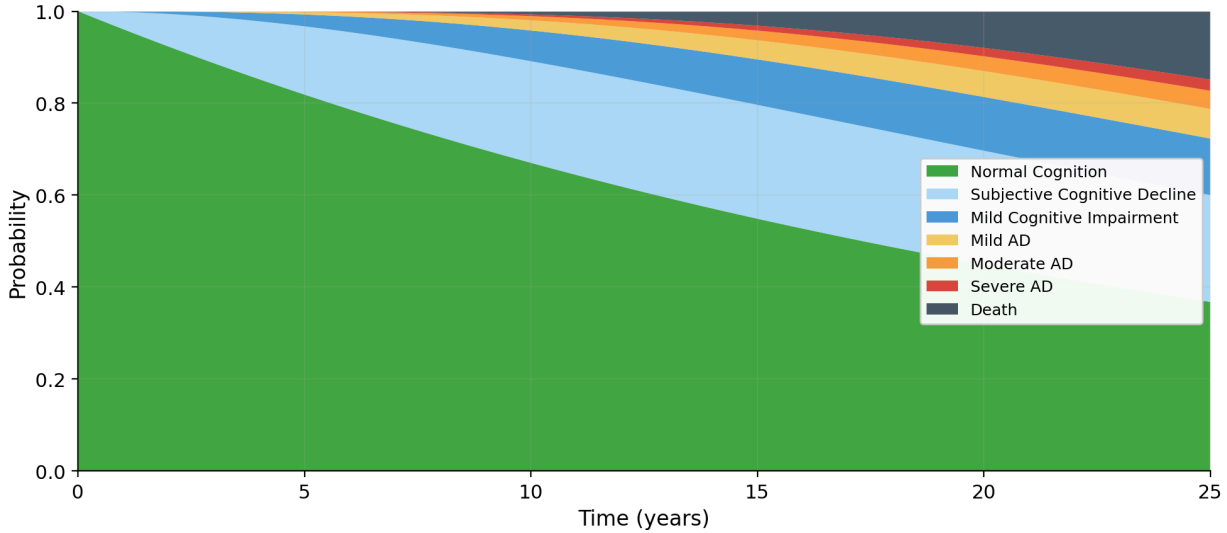
4 European markets, controlled data quality degradation — the same 3 countries I worked in for genomics

MARKET	DATA QUALITY	MEAN DQS	PBW FLAG RATE
Germany (DE)	High (95% coding, 92% labs)	0.85	0.9%
France (FR)	Good (90% coding, 88% labs)	0.80	1.0%
Spain (ES)	Medium (80% coding, 75% labs)	0.70	1.3%
International	Low (60% coding, 50% labs)	0.45	**2.1%**

> The worst market has **2.4x the mispricing risk** — and standard metrics don't see it.

I added Alzheimer's disease in one config file to prove the framework generalises

7 states from normal cognition to death — all transition rates verified against PubMed systematic reviews



Added as configuration: 8 ICD-10 codes, 4 biomarkers (MMSE, MoCA, CSF), 4 medications, ApoE4 risk modifier.

Zero changes to the core engine.

AI-driven underwriting solves five problems current methods cannot

WHAT YOU NEED	RULES / ACTUARIAL	BASIC ML	WHAT I BUILD
Know when to trust the AI	No	No	*DQS + P(wrong) per case*
Handle incomplete records	Reject	Impute (creates PBW)	*Route to right model*
Explain each decision	No	Limited	*SHAP per patient + audit*
Model disease progression	Static tables	No	*CTMC for any disease*
EU AI Act compliance	Partial	Difficult	*Built in from day 1*

Phase 3 vision: LLM agents extract structured data from doctor notes across languages, generate natural language risk explanations, and continuously verify parameters against new literature.

This is how I work: one person + Claude = output of a 5-person team

3 years of Claude evaluation taught me exactly where AI works and where it needs human expertise

MY PROCESS

1. **I define the problem** — PBW in underwriting, from my own published research
2. **Claude builds the code** — pipeline, models, tests, app: 21,000 lines in 48 hours
3. **I verify the science** — LLM agents query PubMed, I validate against domain knowledge
4. **Claude handles scale** — 231 tests, 4 markets, 2 diseases, fact-checked, EU AI Act ready
5. **I make the decisions** — architecture, clinical thresholds, what to show, what matters

> The bottleneck is no longer engineering capacity. It's **domain expertise and data access**.

In 90 days I'd deliver calibrated PBW detection on real Allianz data

MONTH 1 — AUDIT & MAP

Audit current underwriting pipeline for PBW risk. Map data quality across Allianz markets. Identify the **3 highest-impact failure modes**.

MONTH 2 — CALIBRATE & TRAIN

Calibrate DQS on real Allianz data. Retrain model router on actual data profiles. Deliver **first real-data PBW prevalence estimate** by market.

MONTH 3 — SHIP & DOCUMENT

Production prototype: **REST API** for per-case quality scoring. Validated PBW detection rates. **EU AI Act compliance documentation**.

> “I built this proof of concept in 48 hours. Imagine what I do with real data and 90 days.”

Let's build this together.

Tim Reska
PhD Helmholtz Munich · 8 publications · 3 years evaluating Claude

timreska@gmail.com · github.com/ttmgr · linkedin.com/in/timreska

All data in this demo is synthetic. The system, the science, and the skills are real.